Time Series Analysis

PARAMI UNIVERSITY

Time Series Analysis

Today's Topic: Autoregressive Model

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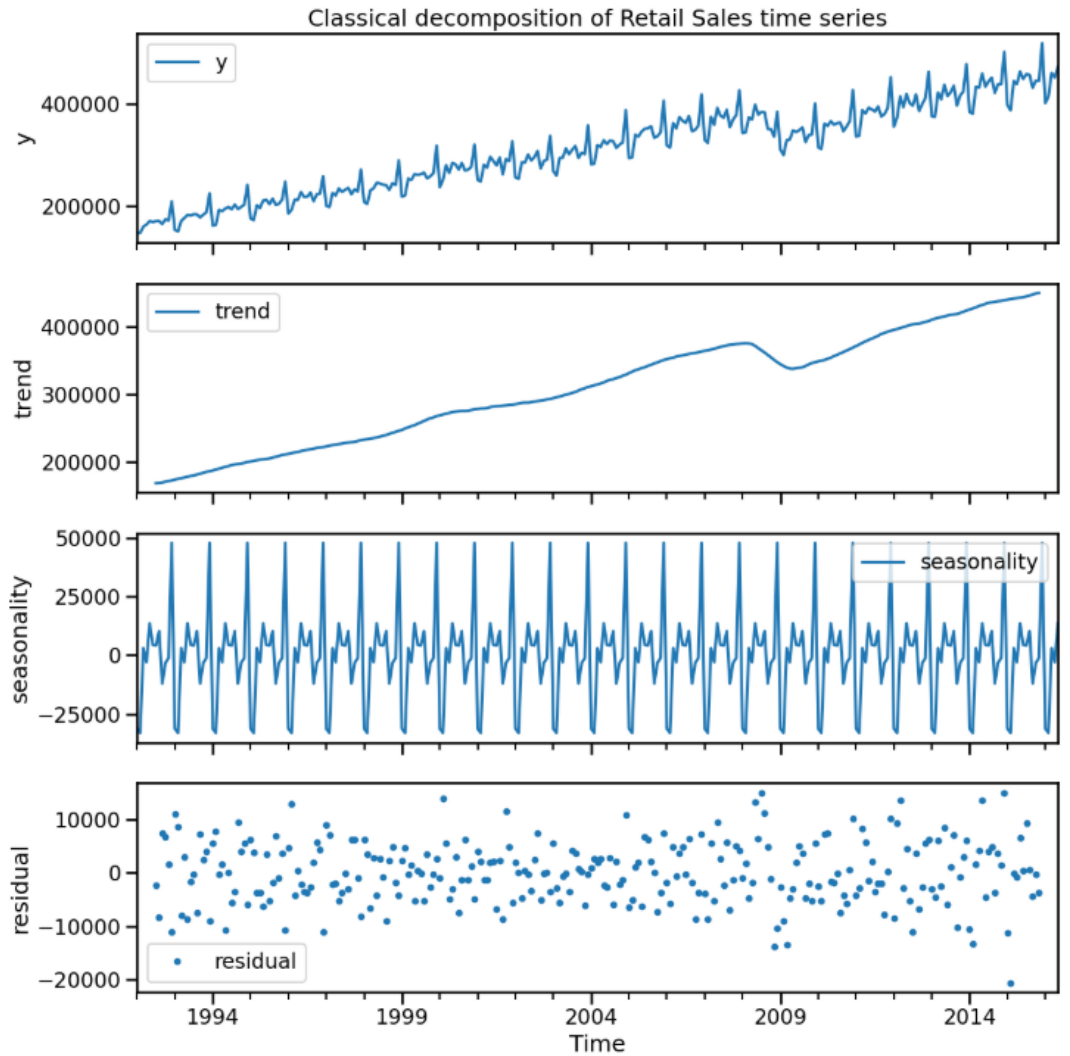
https://sithuaung-ink.github.io/data-analyst-portfolio/

2/25/2026

Motivation

- Three properties already covered, which are trend, seasonality, and white noise.
- A new property of a time series will be introduced today:
  - Autoregressive (AR) property

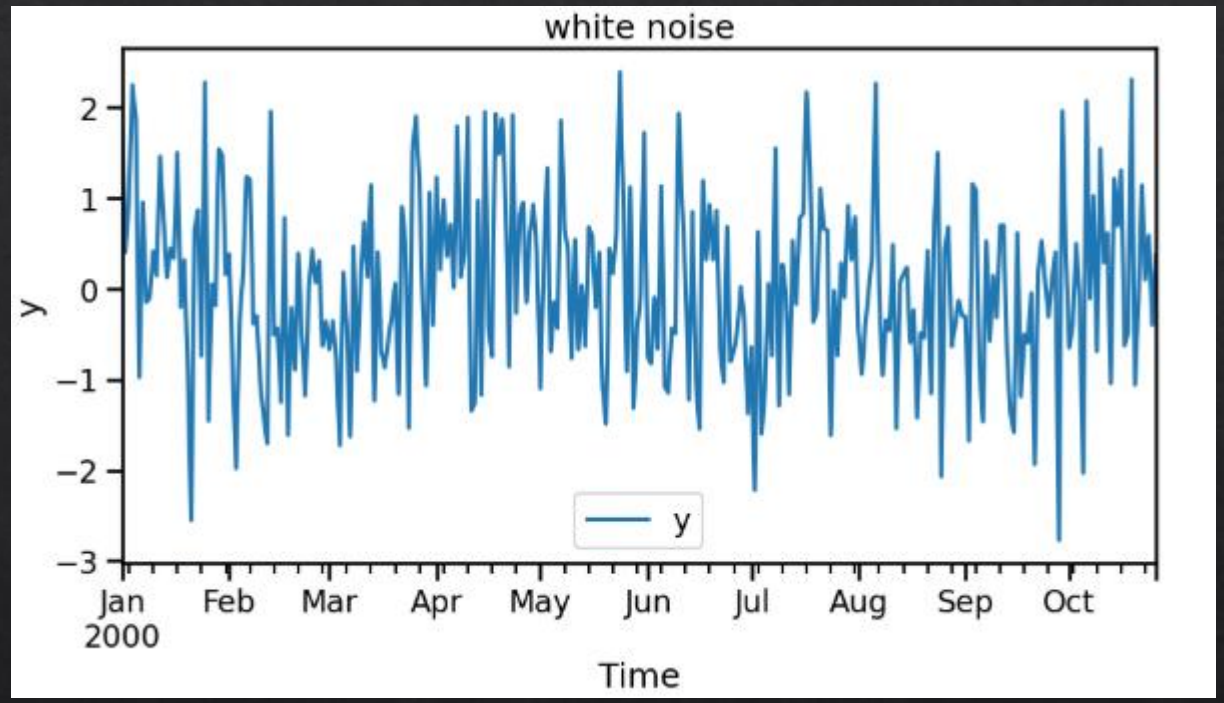


Scope

- AR processes are a large topic covered in the theory of time series analysis and are a broader topic than just forecasting.
- Discuss the definitions of an AR process.
- Provide an intuition about the behavior of the AR process.

White Noise

- White noise has no predictive information in past values, as there is no correlation between any two points in time.



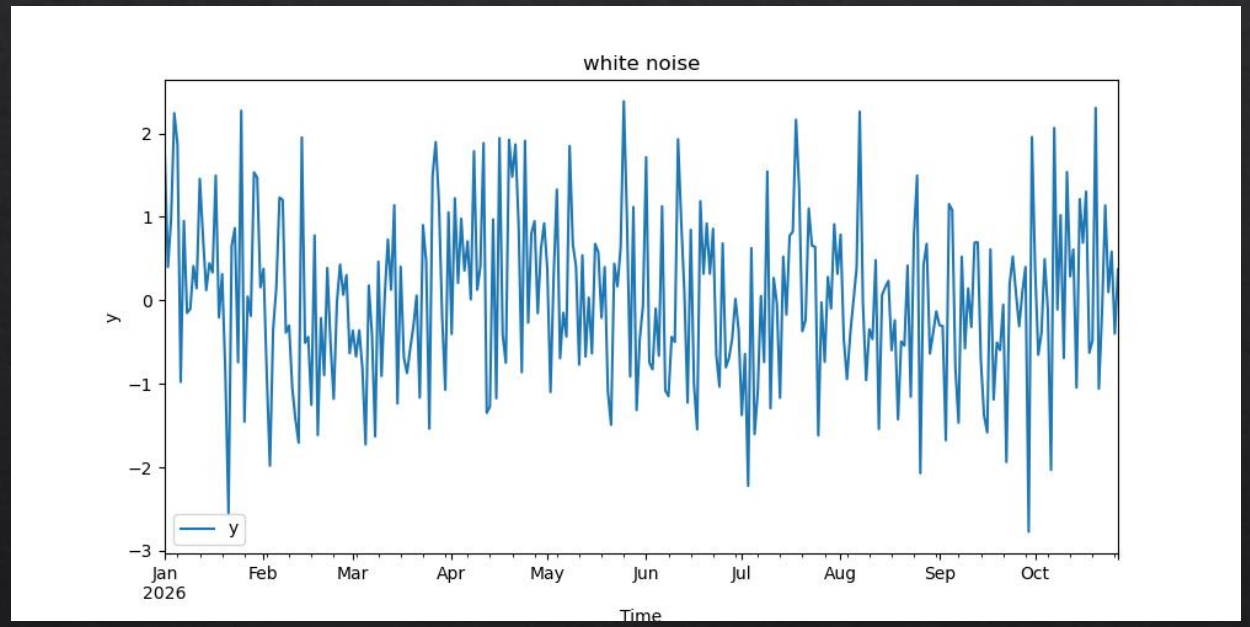
Autoregressive (AR) process

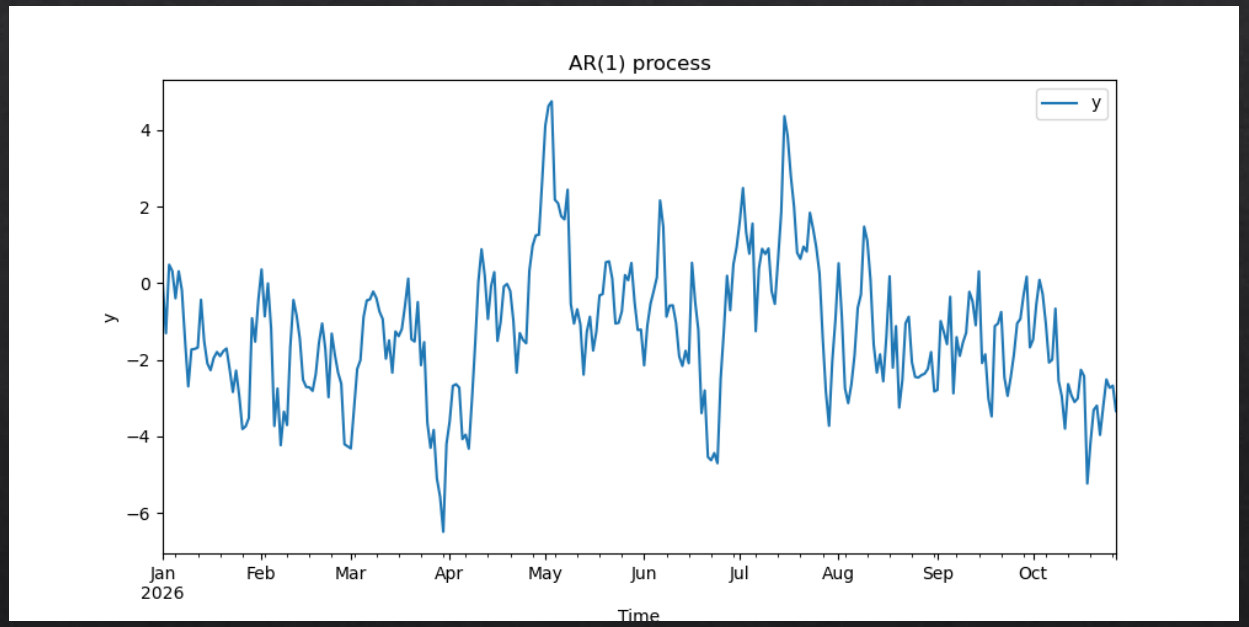
- Let's generate a time series y_t which is determined only from the previous value y_{t-1}, a constant and some white noise ϵ_t .

$$y_t = c + \varphi_1 y_{t-1} + \epsilon_t$$

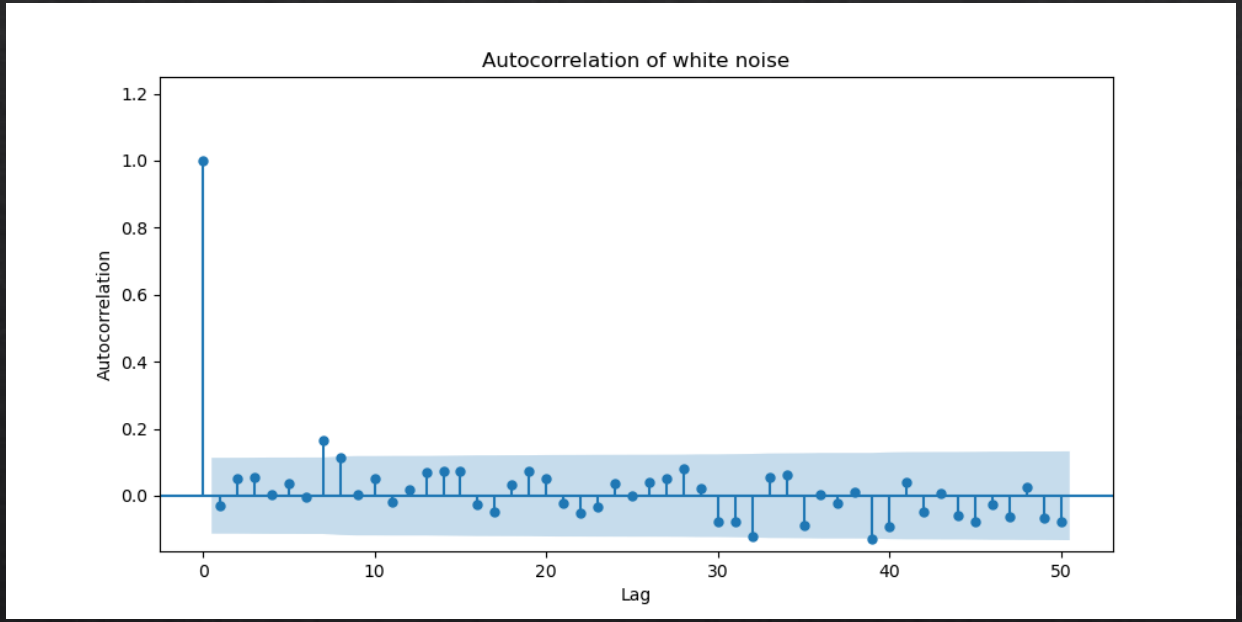
- Where $\varphi_1, \varphi_2, \varphi_3, \dots, \varphi_p$ are real-valued parameter with $\varphi_p \neq 0$.
- Simulate the AR(1) process with Python code.

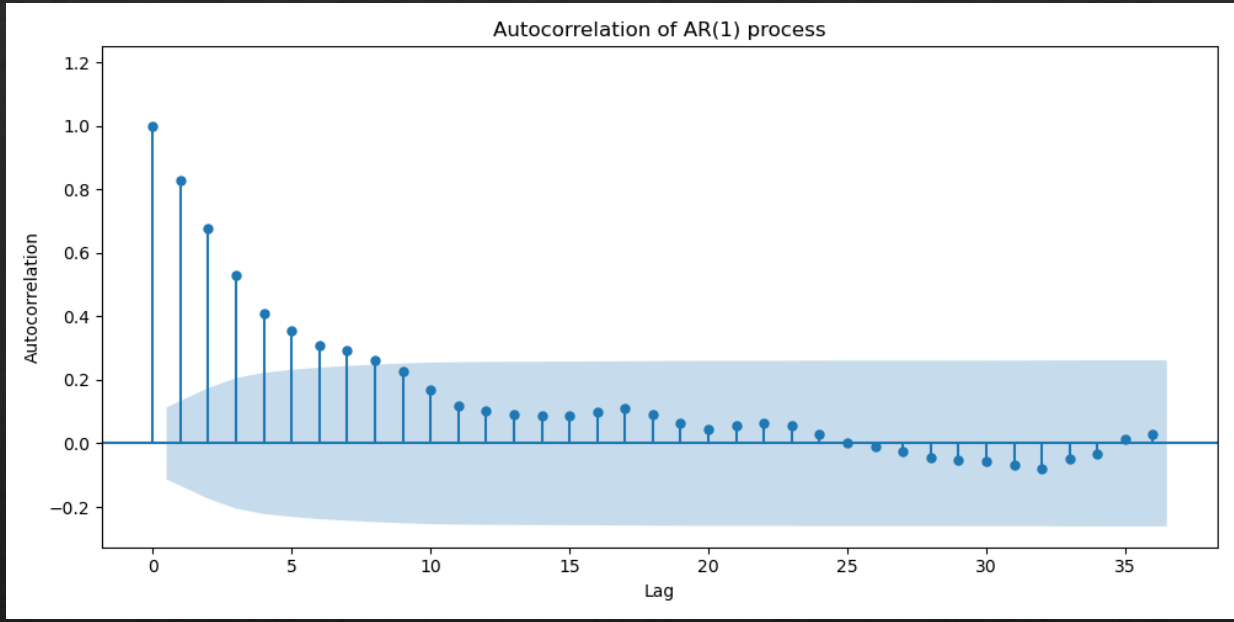
Autoregressive (AR) process





Autoregressive (AR) process





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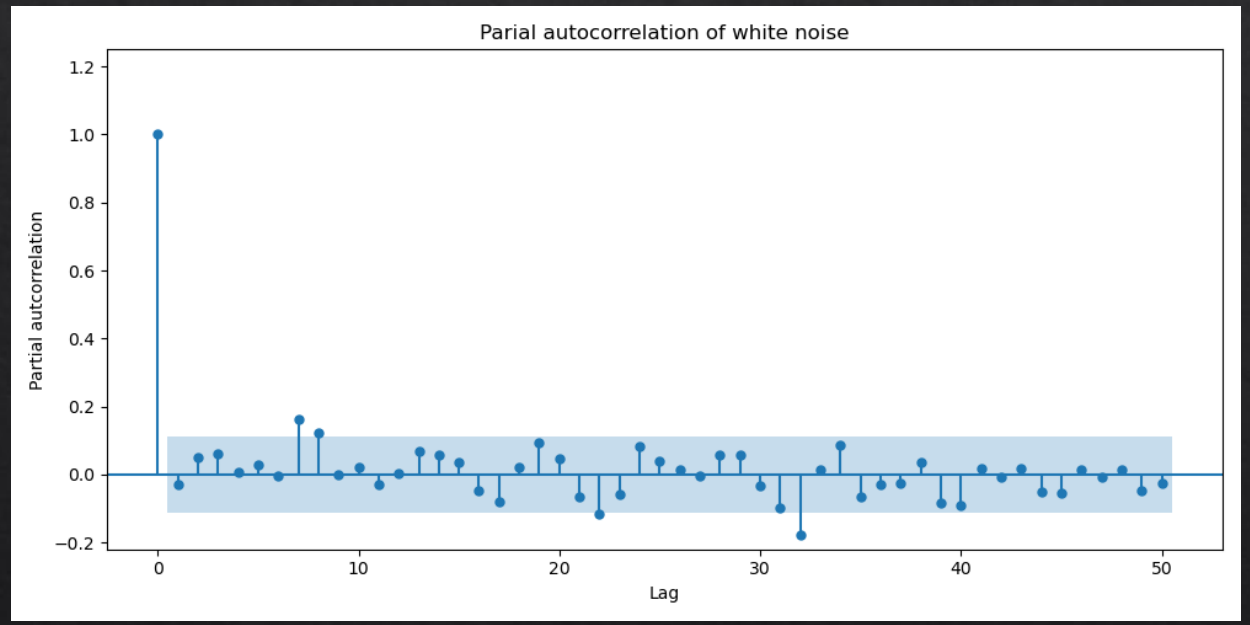
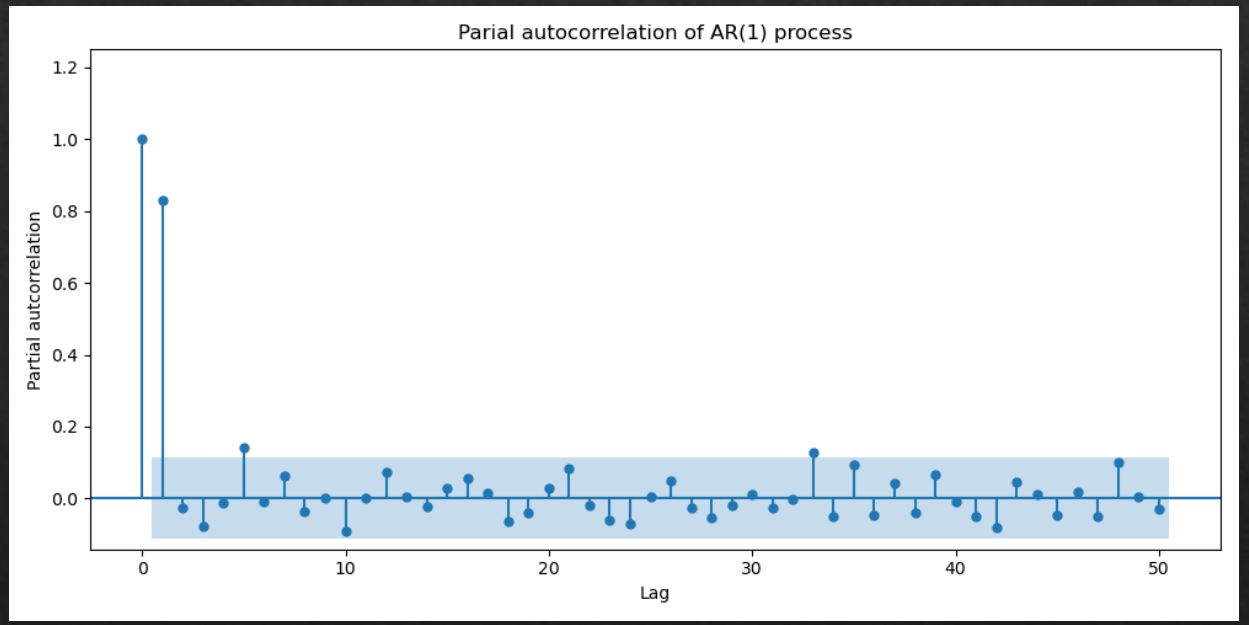
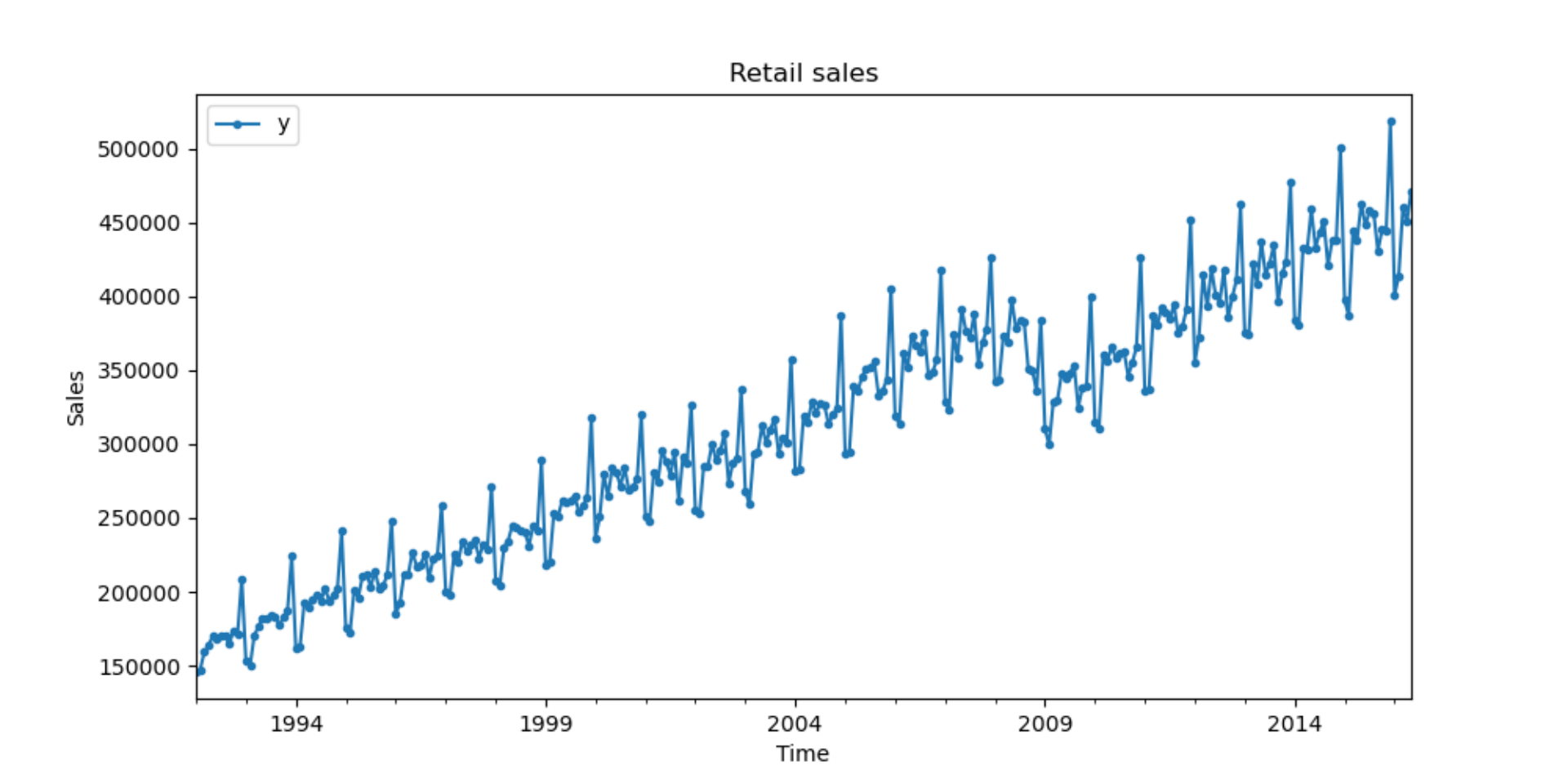




Retail Sales - Lag Feature



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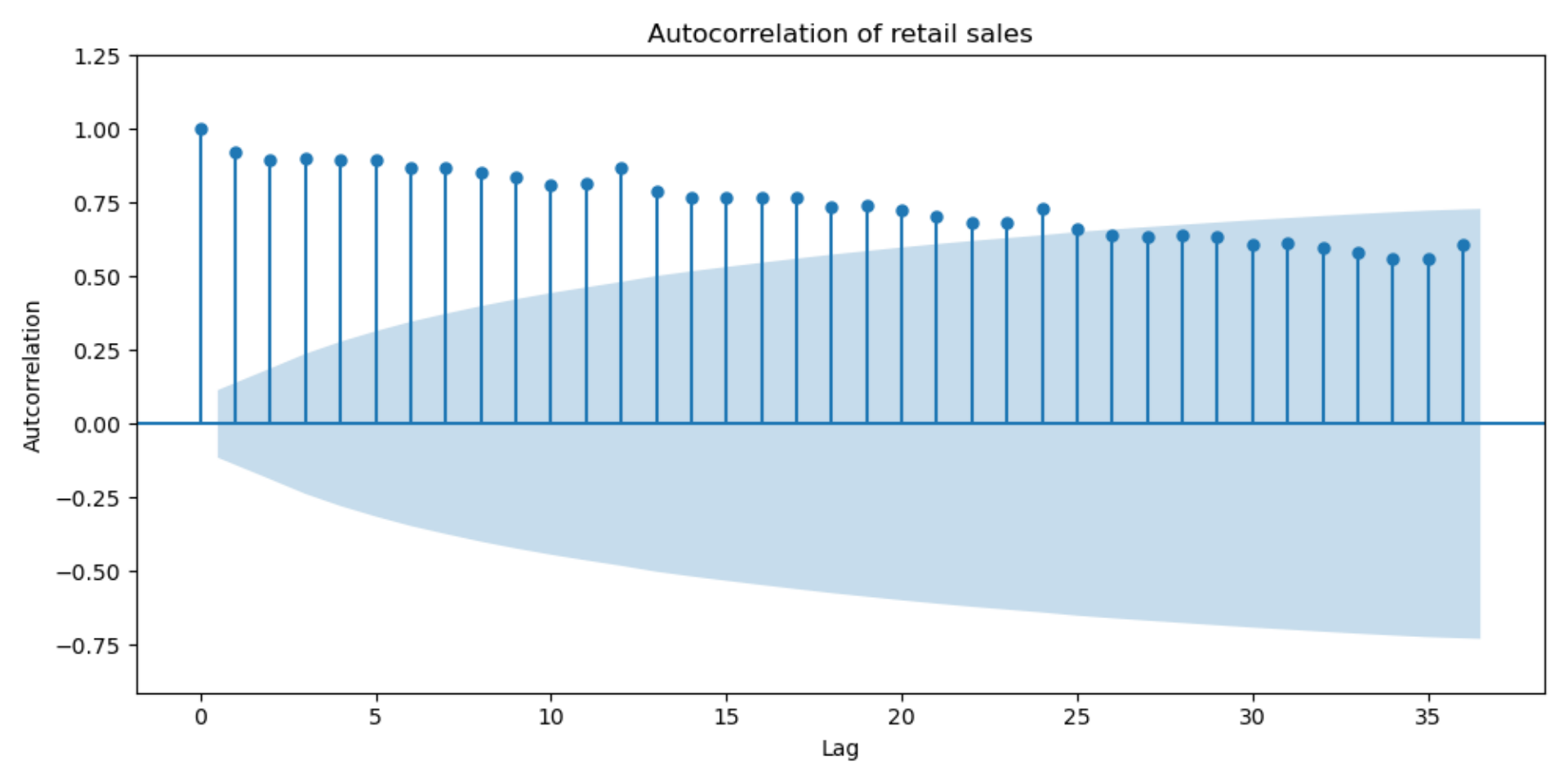

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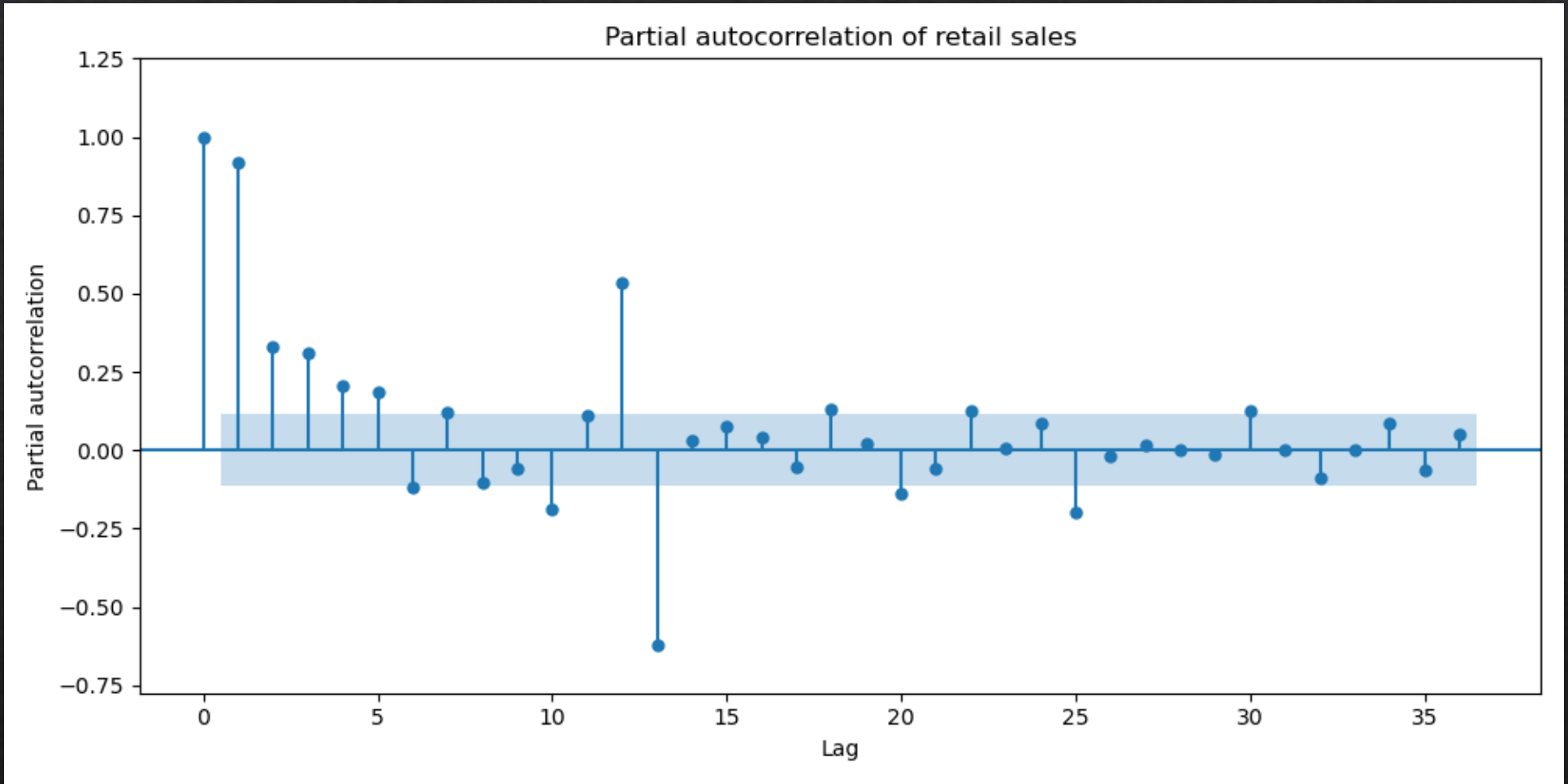

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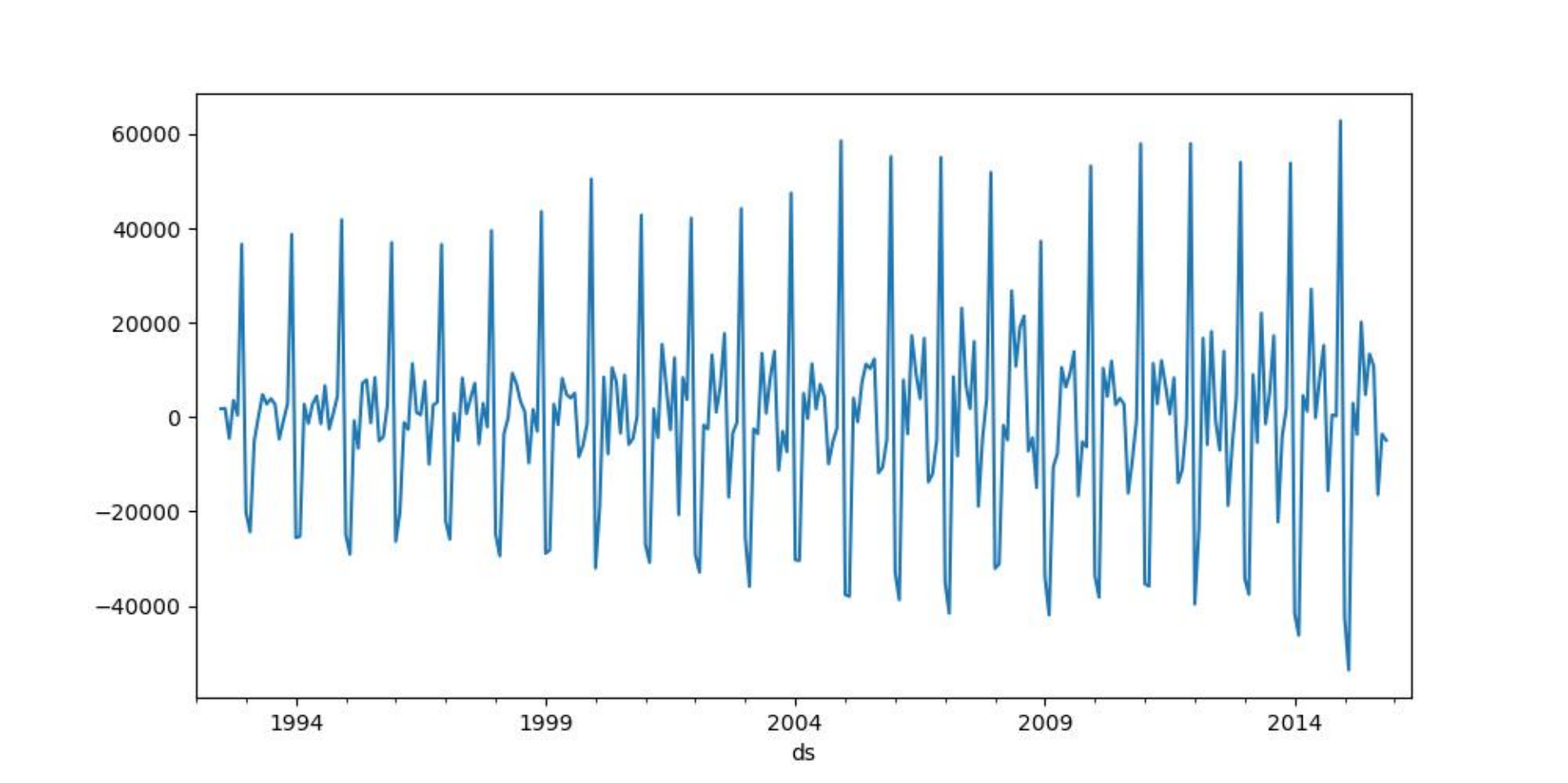

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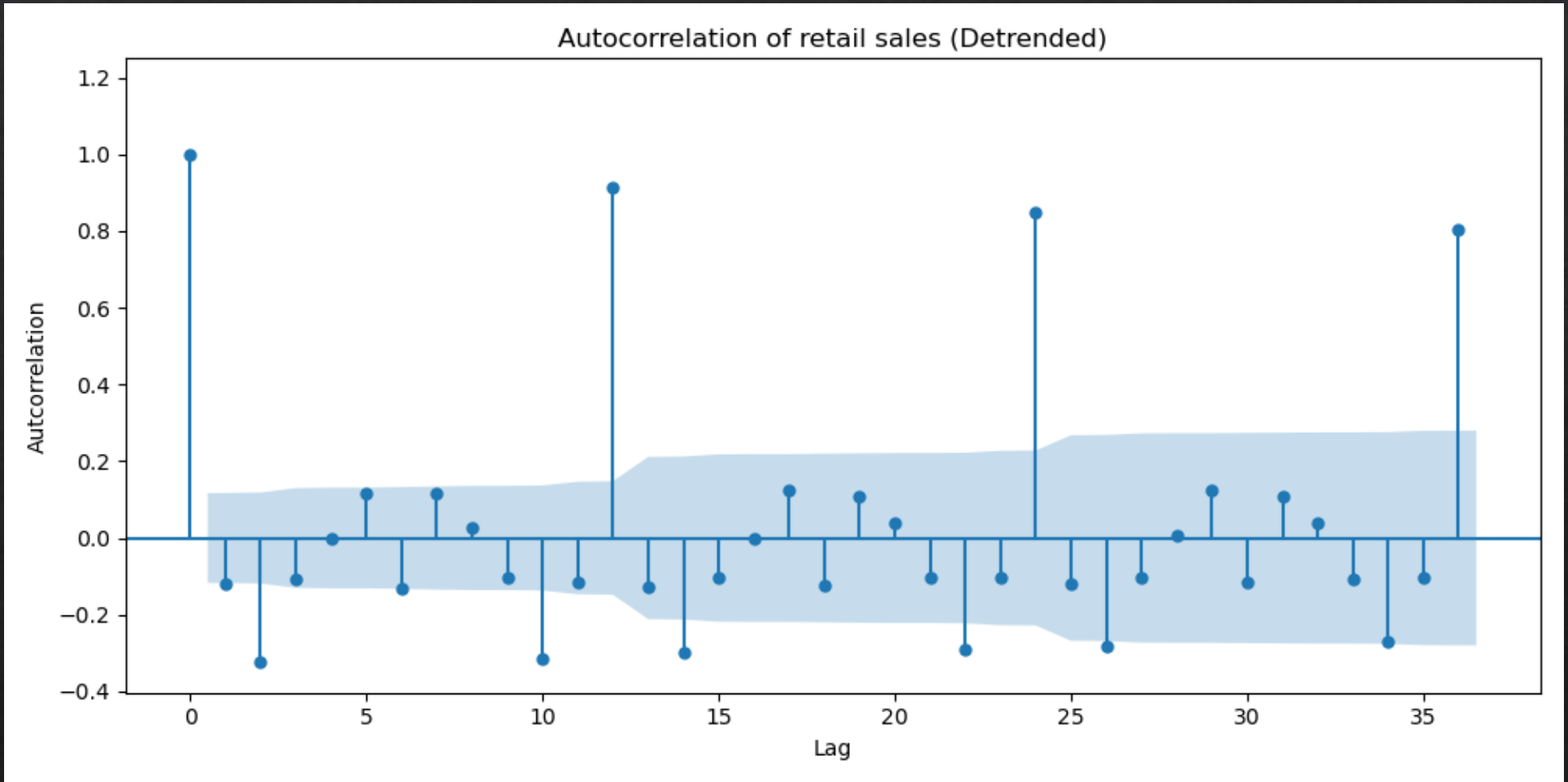

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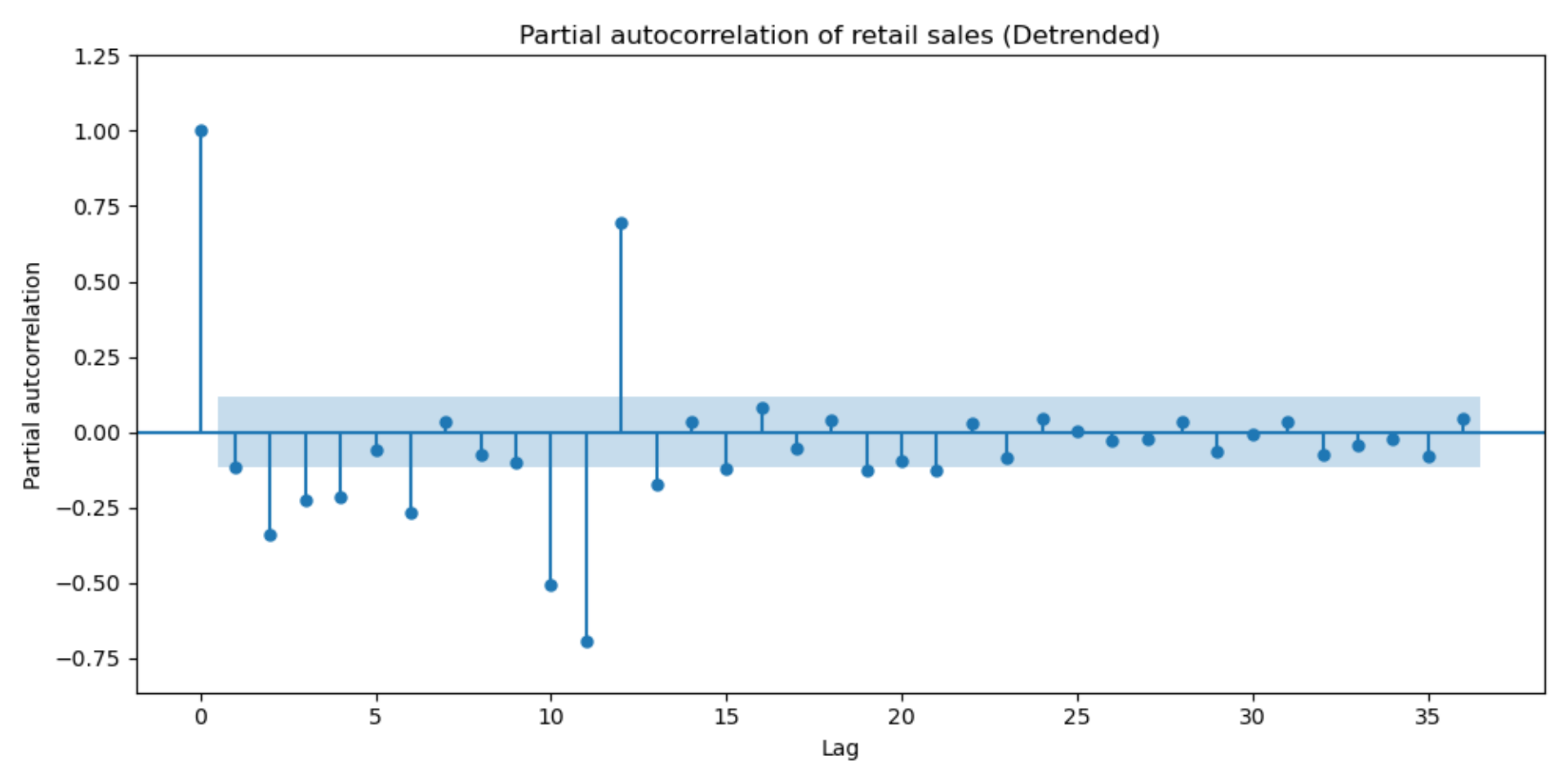

Retail Sales - Lag Feature

- Create the lag feature based on the PACF plot.
- y_t and y_{t-12}

ds	y	y_lag_12
1992-01-01 00:00:00	146376	nan
1992-02-01 00:00:00	147079	nan
1992-03-01 00:00:00	159336	nan
1992-04-01 00:00:00	163669	nan
1992-05-01 00:00:00	170068	nan
1992-06-01 00:00:00	168663	nan
1992-07-01 00:00:00	169890	nan
1992-08-01 00:00:00	170364	nan
1992-09-01 00:00:00	164617	nan
1992-10-01 00:00:00	173655	nan
1992-11-01 00:00:00	171547	nan
1992-12-01 00:00:00	208838	nan
1993-01-01 00:00:00	153221	146376
1993-02-01 00:00:00	150087	147079
1993-03-01 00:00:00	170439	159336
1993-04-01 00:00:00	176456	163669
1993-05-01 00:00:00	182231	170068
1993-06-01 00:00:00	181535	168663
1993-07-01 00:00:00	183682	169890
1993-08-01 00:00:00	183318	170364
1993-09-01 00:00:00	177406	164617
1993-10-01 00:00:00	182737	173655